

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705C - LIMIT STATE DESIGN OF STEEL STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

1. _____ is a disadvantage of Steel.
a. High durability b. Reusable
c. High strength per unit mass d. Fire and corrosion resistance
2. A column that can support same load in compression as it can in tension is called _____
a. intermediate column b. long column
c. short column d. cannot be determined
3. A tension member in which reversal of direct stress occurs due to loads other than wind or seismic forces, the maximum effective slenderness ratio is _____
a. 180 b. 250 c. 300 d. 350
4. The design nominal strength of fillet weld is given by _____
a. f_u b. $\sqrt{3} f_u$ c. $f_u/\sqrt{3}$ d. $f_u/(1.25 \times \sqrt{3})$
5. Grade 4.6 bolt has nominal ultimate stress of _____
a. 460Mpa b. 400Mpa c. 600Mpa d. None of these
6. For the calculation of net area of flat with staggered bolts, the area to be deducted from gross area is _____
a. nd b. $n'p2t/8g$ c. $ndt - n'p2t/4g$ d. $nd + n'p2t/4g$
7. A tension member in which reversal of direct stress occurs due to loads other than wind or seismic forces, the maximum effective slenderness ratio is _____
a. 180 b. 250 c. 300 d. 350
8. _____ is the effective length when one end of compression member is fixed and other end is free.
a. $0.65L$ b. $0.8L$ c. L d. $2L$
9. Effective slenderness ratio of laced column shall be _____
a. equal to the maximum slenderness ratio of column
b. 1.05 times the maximum slenderness ratio of column
c. 0.5 times the maximum slenderness ratio of column
d. 2 times the maximum slenderness ratio of column
10. Minimum radius of gyration for lacing flats is _____
a. $t/\sqrt{12}$ b. $t/12$ c. $t/\sqrt{24}$ d. $t/24$
11. The members of the roof truss that support the purlins are known as _____
a. Principal rafters b. Principal tie c. Main strut d. Main tie

- 12 The space between two adjacent trusses is called ____
a. Purlin b. Bay c. knee d. Braces

Q2. Solve the following.

- A) Explain why steel is best suited construction material for the following: 6
a. Large span industrial truss
b. Buildings to be shifted
c. Hilly areas where construction is possible in summer.
- B) Explain the advantages and disadvantages of using steel structures. 6

Q3. Solve the following.

- A) Explain the following terms: 6
a. Pitch of the bolts
b. Gauge distance
c. Edge Distance
- B) Determine the bolt value of 20 mm diameter bolt connecting 10 mm plate in: 6
a) single shear
b) double shear.
Bolt used are 4.6 grade and plate of 410.

Q4. Solve Any Two of the following.

- A) A single angle discontinuous member ISA 110 X 110 X10mm with single bolted connection is 2.5m long. Calculate the design compressive strength of the section if it is connected by one bolt at each end. Take $F_y=250$, $r_{vv}=21.4\text{mm}$, A_g of ISA=110 X 110 X 10mm=2106mm². 6
- B) Explain Lug angle and shear leg with neat sketch. 6
- C) Explain the failure modes of axially loaded column in detail. 6

Q5. Solve Any Two of the following.

- A) Design a single lacing system for a column composed of 2ISMC 300@ 35.8 Kg/m placed back to back at clear spacing of 200mm. Axial factored load on column is 1500KN. Effective length of column is 5m. 6
- B) Explain lacing and battening in detail with neat sketch. 6
- C) Explain with neat sketches 6
1) Slab base 2) Gusseted Base 3) Grillage Foundation

Q6. Solve Any Two of the following.

- A) A roof truss shed is to be built in Mumbai for an industry. The size of the shed is 6 24m X 6 40m. The height of the building is 12m at the eaves. Determine the basic wind pressure. 6
- B) Describe the process of design of truss with details of 6
1) Wind load calculations 2) different load combinations
- C) Explain factors affecting plastic moment capacity in detail. 6

*** End ***